

Template

Studentnames and studentnumbers here

2026-06-16

Set-up your environment

Title Page

Include your names

Include the tutorial group number

Include your tutorial lecturer's name

Part 1 - Identify a Social Problem

Use APA referencing throughout your document. Here's a link to some explanation.

1.1 Describe the Social Problem

Include the following:

- Why is this relevant?
- ...

Part 2 - Data Sourcing

2.1 Load in the data

Preferably from a URL, but if not, make sure to download the data and store it in a shared location that you can load the data in from. Do not store the data in a folder you include in the Github repository!

```
#loading in the datasets  
total_govt_exp <- read_csv("data/total_govt_exp.csv")
```

```
## Rows: 340 Columns: 23  
## -- Column specification -----  
## Delimiter: ","  
## chr (17): STRUCTURE, STRUCTURE_ID, STRUCTURE_NAME, freq, Time frequency, uni...  
## dbl (2): TIME_PERIOD, OBS_VALUE
```

```
## lgl (4): Time, Observation value, CONF_STATUS, Confidentiality status (flag)
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
general_income_ineq_gini_ <- read_csv("data/general_income_ineq(gini).csv")
```

```
## Rows: 368 Columns: 19
## -- Column specification -----
## Delimiter: ","
## chr (13): STRUCTURE, STRUCTURE_ID, STRUCTURE_NAME, freq, Time frequency, age...
## dbl (2): TIME_PERIOD, OBS_VALUE
## lgl (4): Time, Observation value, CONF_STATUS, Confidentiality status (flag)
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
tertiary_edu_govt_exp <- read_csv("data/tertiary_edu_govt_exp.csv")
```

```
## Rows: 320 Columns: 25
## -- Column specification -----
## Delimiter: ","
## chr (19): STRUCTURE, STRUCTURE_ID, STRUCTURE_NAME, freq, Time frequency, uni...
## dbl (2): TIME_PERIOD, OBS_VALUE
## lgl (4): Time, Observation value, CONF_STATUS, Confidentiality status (flag)
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

midwest is an example dataset included in the tidyverse package

midwest is an example dataset included in the tidyverse package

2.2 Provide a short summary of the dataset(s)

In this case we see 28 variables, but we miss some information on what units they are in. We also don't know anything about the year/moment in which this data has been captured.

These are things that are usually included in the metadata of the dataset. For your project, you need to provide us with the information from your metadata that we need to understand your dataset of choice.

2.3 Describe the type of variables included

Think of things like:

- Do the variables contain health information or SES information?
- Have they been measured by interviewing individuals or is the data coming from administrative sources?

For the sake of this example, I will continue with the assignment...

Part 3 - Quantifying

3.1 Data cleaning

Say we want to include only larger distances (above 2) in our dataset, we can filter for this.

```
#inladen tidyverse
library(tidyverse)

#income_ineq aan het selecteren
income_ineq <- read.csv("data/general_income_ineq(gini).csv") %>%
  select(geo, Geopolitical.entity..reporting., TIME_PERIOD, OBS_VALUE) %>%
  rename(
    country = Geopolitical.entity..reporting.,
    year = TIME_PERIOD,
    gini = OBS_VALUE) %>%
  filter(
    geo != "EA20",
    geo != "EU27_2020",
    geo != "Kosovo*")
```

```
education <- read.csv("data/tertiary_edu_govt_exp.csv")
education <- education[, c("geo", "Geopolitical.entity..reporting.", "TIME_PERIOD", "OBS_VALUE")]
names(education)[names(education) == "Geopolitical.entity..reporting."] <- "country"
names(education)[names(education) == "TIME_PERIOD"] <- "year"
names(education)[names(education) == "OBS_VALUE"] <- "edu_exp"
education <- education[-(109:119), ]
```

```
tot_gov <- read.csv("data/total_govt_exp.csv")
tot_gov <- tot_gov[, c("geo", "Geopolitical.entity..reporting.", "TIME_PERIOD", "OBS_VALUE")]
names(tot_gov)[names(tot_gov) == "Geopolitical.entity..reporting."] <- "country"
names(tot_gov)[names(tot_gov) == "TIME_PERIOD"] <- "year"
names(tot_gov)[names(tot_gov) == "OBS_VALUE"] <- "tot_exp"
tot_gov <- tot_gov[-(141:150), ]
tot_gov <- tot_gov[-(81:110), ]
```

Please use a separate ‘R block’ of code for each type of cleaning. So, e.g. one for missing values, a new one for removing unnecessary variables etc.

3.2 Generate necessary variables

Variable 1

```
library(tidyverse)
education$country <- NULL
rel_edu <- education %>% inner_join(tot_gov, by = c("geo", "year"))
rel_edu <- rel_edu %>% mutate(education_share = edu_exp / tot_exp * 100)
rel_edu$education_share <- round(rel_edu$education_share, 4)
avg_rel_edu <- rel_edu %>% group_by(geo, country) %>% summarise(avg_edu = mean(education_share), n_obs = n())
```

‘summarise()’ has regrouped the output.

```
## i Summaries were computed grouped by geo and country.
## i Output is grouped by geo.
## i Use 'summarise(.groups = "drop_last")' to silence this message.
## i Use 'summarise(.by = c(geo, country))' for per-operation grouping
## ('?dplyr::dplyr_by') instead.
```

Variable 2

```
#create dataset w gini avg (per country)
avg_gini_data <- income_ineq %>% group_by(geo, country) %>%
summarise (avg_gini = mean(gini), n_obs = n()) %>%
arrange(desc(avg_gini))
```

```
## 'summarise()' has regrouped the output.
## i Summaries were computed grouped by geo and country.
## i Output is grouped by geo.
## i Use 'summarise(.groups = "drop_last")' to silence this message.
## i Use 'summarise(.by = c(geo, country))' for per-operation grouping
## ('?dplyr::dplyr_by') instead.
```

```
gini_edu_data <- full_join(avg_gini_data, avg_rel_edu, by = c("geo", "country"))
gini_edu_data$n_obs.x <- NULL
gini_edu_data$n_obs.y <- NULL
```

```
gini_edu_data$region <- replace(gini_edu_data$geo, gini_edu_data$geo %in% c("DK", "EE", "FI", "IE", "IS",
gini_edu_data$region <- replace(gini_edu_data$region, gini_edu_data$region %in% c("AT", "BE", "CH", "DE",
gini_edu_data$region <- replace(gini_edu_data$region, gini_edu_data$region %in% c("AL", "CY", "EL", "ES",
gini_edu_data$region <- replace(gini_edu_data$region, gini_edu_data$region %in% c("BG", "CZ", "HU", "PL",
```

Variable 3

```
year_avg_exp_eu <- rel_edu %>% group_by(year) %>%
summarise(eu_ave_exp = mean(education_share),
aantalLanden = n())
```

Variable 4

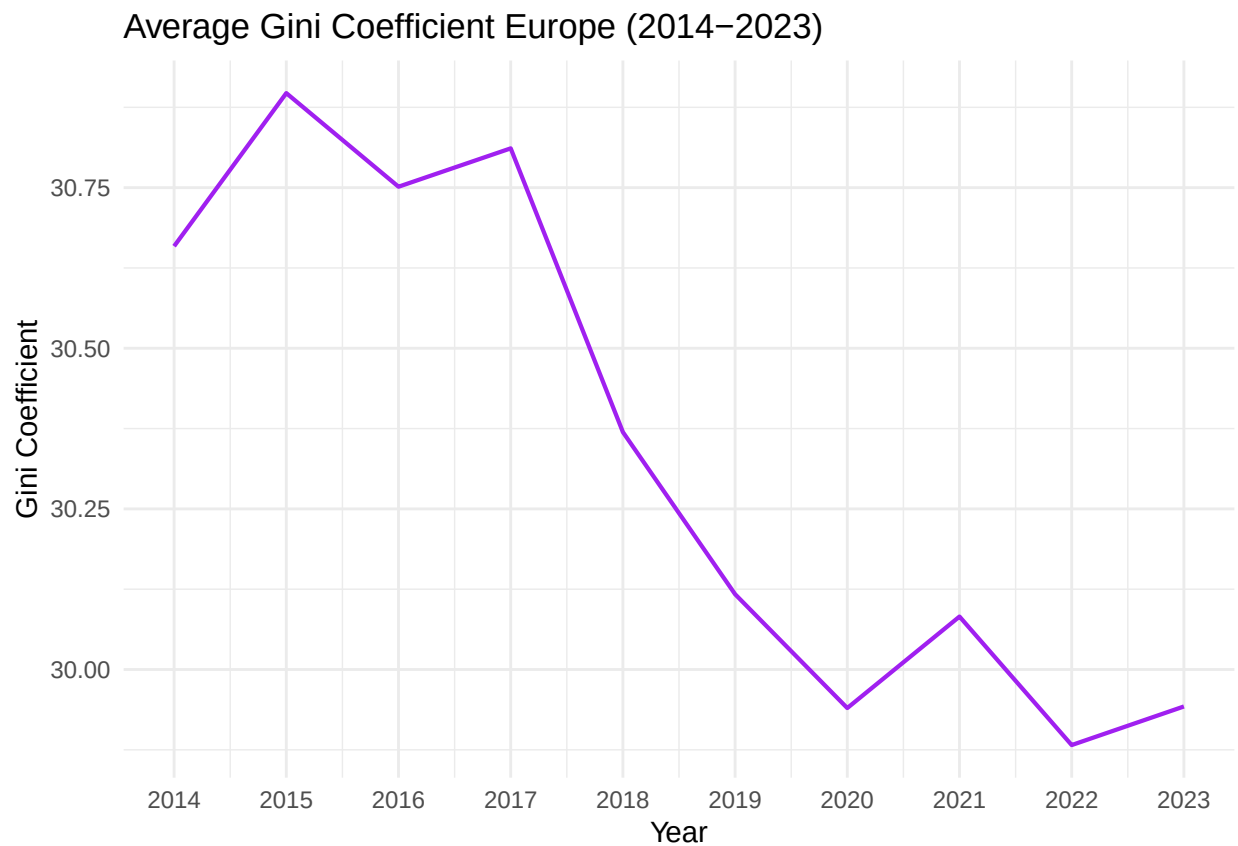
```
gini_avg_pyear_eu <- income_ineq[income_ineq$country != "Kosovo*"] %>% group_by(year) %>%
summarise(avg_gini = mean(gini),
aantalLanden = n())
```

3.3 Visualize temporal variation

```
library(ggplot2)

gini_avg_pyear_eu %>%
```

```
ggplot(aes(x = year, y = avg_gini)) +
  geom_line(
    colour = "purple",
    linewidth = 0.75) +
  scale_x_continuous(
    breaks = seq(2014, 2023, by = 1)) +
  labs(
    title = "Average Gini Coefficient Europe (2014–2023)",
    x = "Year",
    y = "Gini Coefficient"
  ) +
  theme_minimal()
```



Kosovo weggehaald voor de shock, er was maar één meting van 2018

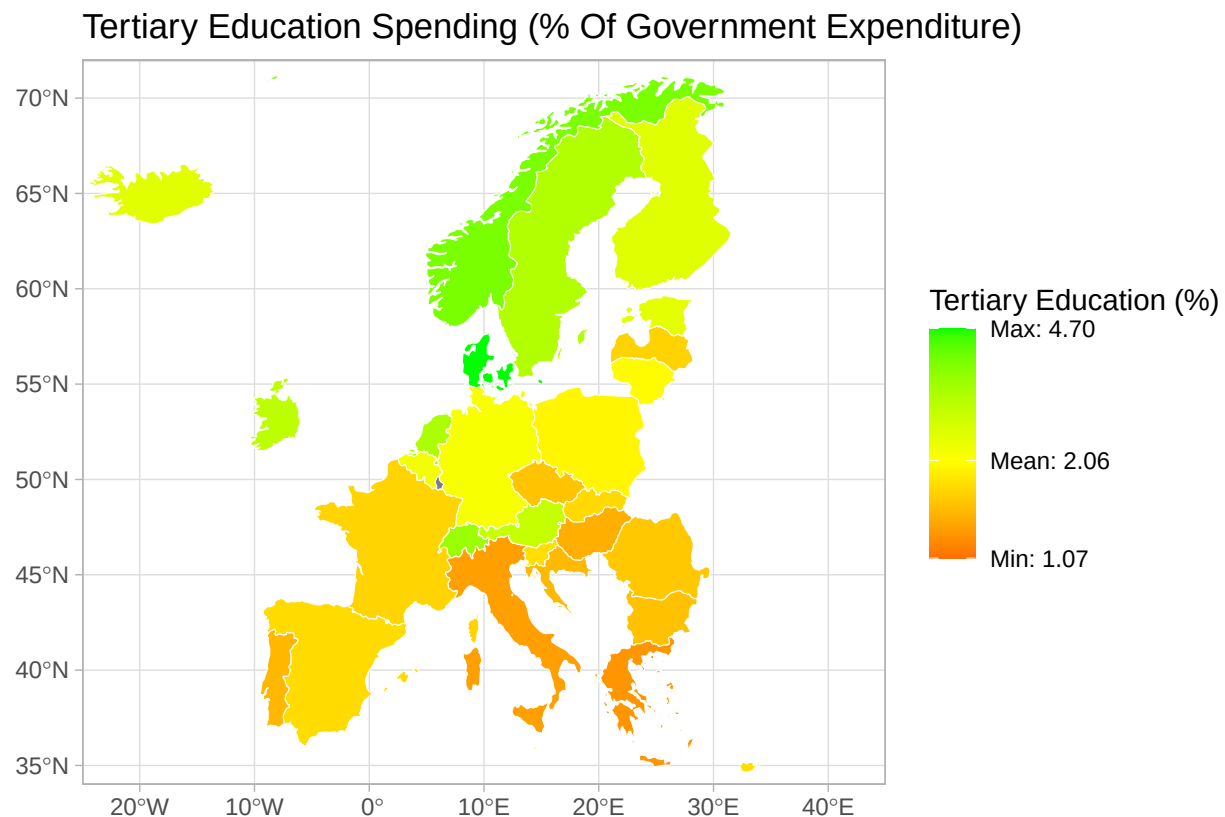
3.4 Visualize spatial variation

```
library(sf)
```

```
## Linking to GEOS 3.14.1, GDAL 3.12.1, PROJ 9.7.1; sf_use_s2() is TRUE
```

```
library(rnaturalearth)

world <- ne_countries(scale = "medium", returnclass = "sf")
map_eu <- world %>%
  inner_join(avg_rel_edu, by = c("name" = "country"))
ggplot(map_eu) +
  geom_sf(aes(fill = avg_edu), color = "white", linewidth = 0.2) +
  scale_fill_gradient2(low = "red", mid = "yellow", high = "green", midpoint = 2.615328, limits = c(1.07, 4.70)) +
  coord_sf(xlim = c(-25, 45), ylim = c(34, 72), expand = FALSE) +
  theme_light() +
  labs(title = "Tertiary Education Spending (% Of Government Expenditure)", fill = "Tertiary Education Spending (% Of Government Expenditure)")
```



Here you provide a description of why the plot above is relevant to your specific social problem.

3.5 Visualize sub-population variation

```
#creating scatterplot from gini_edu_data
ggplot(gini_edu_data,
  aes(x = avg_edu,
    y = avg_gini,
    colour = region)) +
  geom_point() +
  scale_colour_manual(
```

```

values = c(
  "west_europe" = "blue",
  "north_europe" = "orange",
  "south_europe" = "green",
  "east_europe" = "red")) +
scale_x_continuous(
  limits = c(0, 5)) +
scale_y_continuous(
  limits = c(20, 40)) +
labs(
  title = "Relationship Tertiary Education Expenditure and Income Inequality",
  x = "Tertiary Education Spending (%)",
  y = "Gini coefficient",
  colour = "Europe Regions") +
geom_smooth( #regression line
  method = "lm",
  se = FALSE,
  colour = "black",
  linetype = "dotted",
  linewidth = 0.8) + theme_minimal()

```

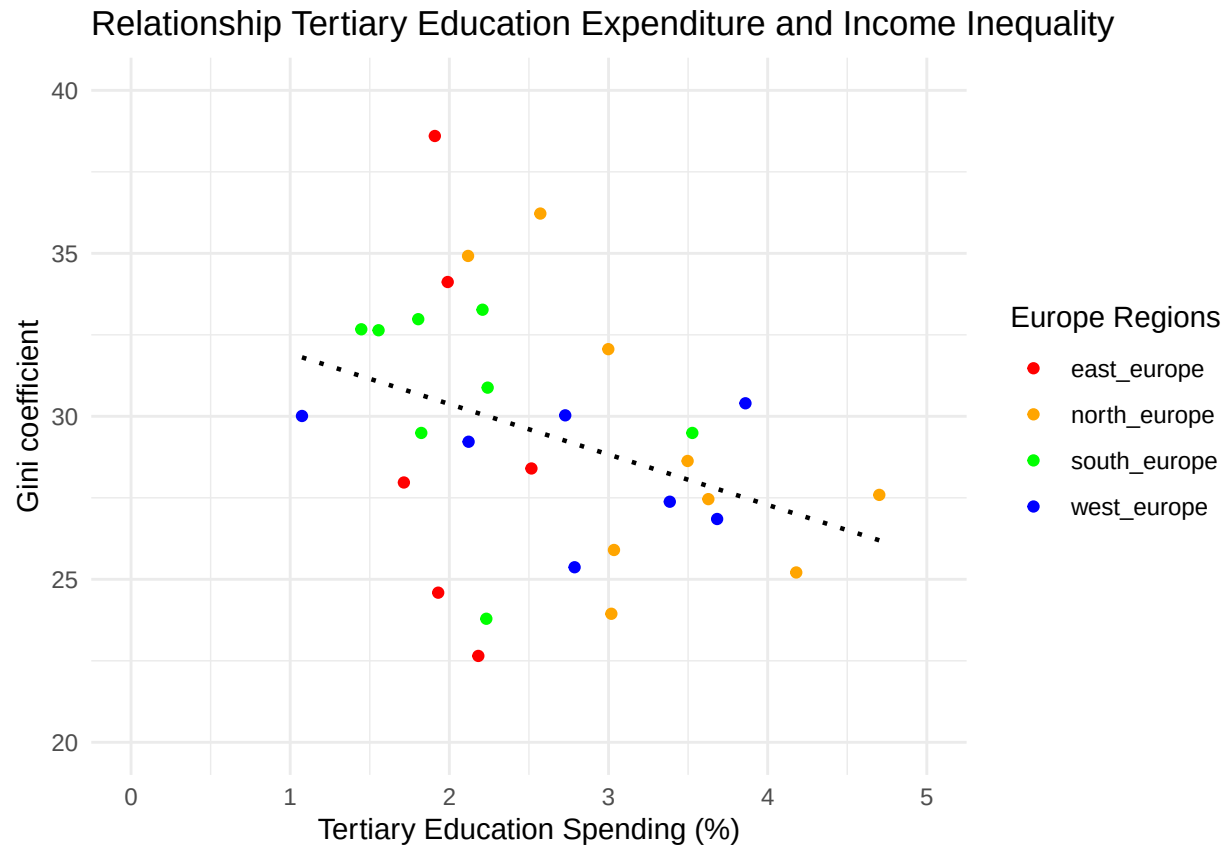
```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 7 rows containing non-finite outside the scale range
```

```
## ('stat_smooth()').
```

```
## Warning: Removed 7 rows containing missing values or values outside the scale range
```

```
## ('geom_point()').
```



What is the poverty rate by state

Here you provide a description of why the plot above is relevant to your specific social problem.

3.6 Event analysis

Analyze the relationship between two variables.

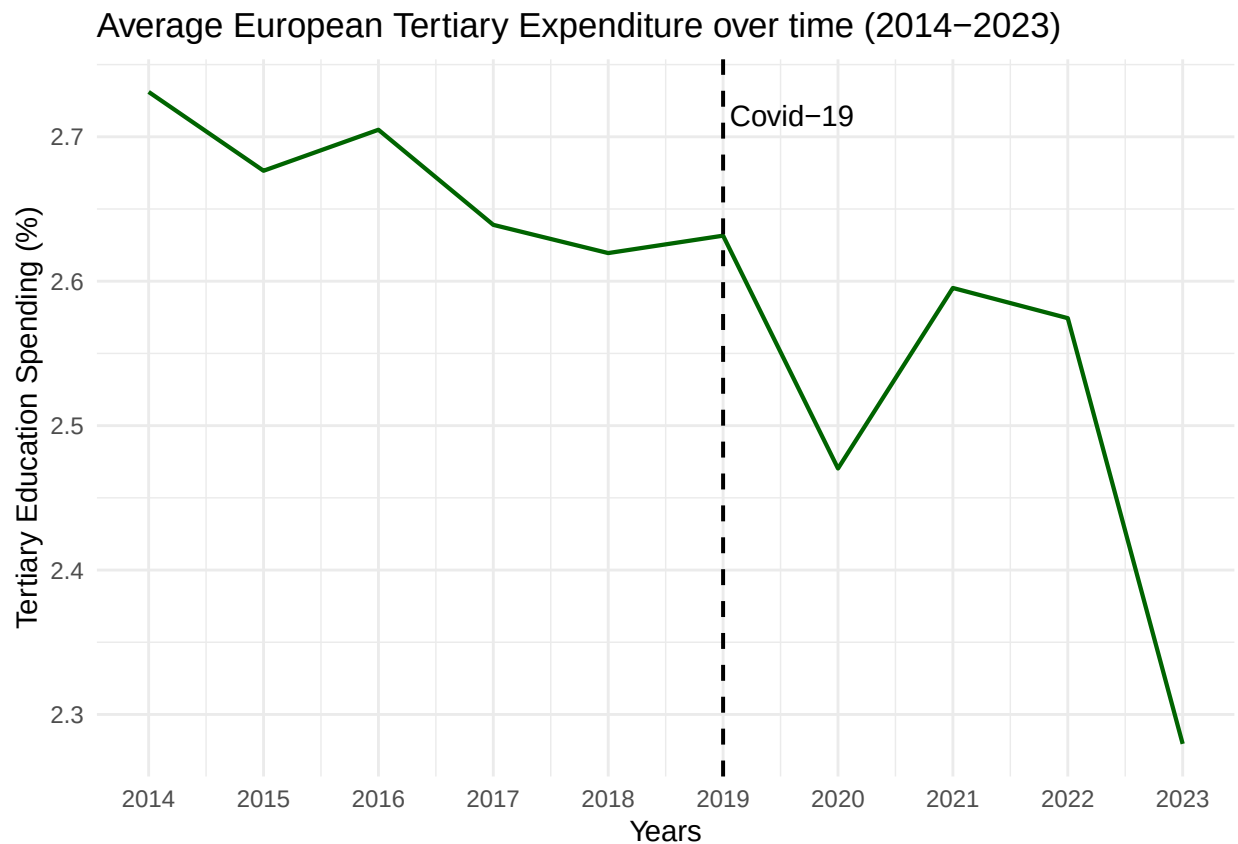
```
#maken van een graafiek met data (government expenditure over tijd)
library(tidyverse)
library(ggplot2)

year_avg_exp_eu %>%
  ggplot(aes(x=year, y=eu_ave_exp)) +
  geom_line(
    colour = "darkgreen",
    linewidth = 0.75) +
  scale_x_continuous(
    breaks = seq(2014,2023, by = 1)) +
  labs(
    title = "Average European Tertiary Expenditure over time (2014-2023)",
    x = "Years",
    y = "Tertiary Education Spending (%)")
  ) +
  theme_minimal()+
  geom_vline(
```

```

xintercept = 2019,
linetype = "dashed",
color = "black",
linewidth = 0.7) +
annotate(
  "text",
  x = 2019.6,
  y = 2.7,
  label = "Covid-19",
  angle = 0,
  vjust = -0.5
)

```



Here you provide a description of why the plot above is relevant to your specific social problem.

Part 4 - Discussion

4.1 Discuss your findings

Part 5 - Reproducibility

5.1 Github repository link

Provide the link to your PUBLIC repository here: ...

5.2 Reference list

Use APA referencing throughout your document.